Math 141, F11
Quiz 0

Name: Answer
SSN: xxx-xx-

Instructions: There are totally 4 problems.

Problem 1. Simplify

$$\frac{1}{3}+\frac{3}{4}=\frac{4}{12}+\frac{9}{12}=\frac{13}{12}$$

Problem 2. Find

$$\sin\frac{\pi}{3}=\frac{\sqrt{3}}{2}$$

$$\tan 45^\circ = 1$$

Problem 3. Simplify

$$x^6\cdot x^{\frac{1}{3}}=x^{6+\frac{1}{3}}=x^{\frac{19}{3}}$$

Problem 4. Solve the following quadratic equation:

$$x^2+2x-3=0.$$

Method 1. Using quadratic formula

$$x=\frac{-2\pm\sqrt{2^2-4\cdot1\cdot(-3)}}{2}$$

$$=\frac{-2\pm\sqrt{16}}{2}$$

$$=-1\pm2$$

$$\Rightarrow x=1 \text{ or } x=-3$$

Method 2. By factorization

$$x^2+2x-3=0$$

$$(x-1)(x+3)=0$$

$$\Rightarrow x=1 \text{ or } x=-3$$